

Identifying Interaction Design and Functional Layout

Practical implementation of interaction design and functional layouts

Case study: **Personal App — Complete Application** (Notes, Habits, Calendar, Calculator, Life Tracker, Settings & Onboarding)

Submitted by: SAI CHANDAN KS (23MIS0115)

Tools: Flutter 3.22+ · Dart 3.7+ · Provider · SQLite

1. Aim

To identify user requirements for a multi-feature personal application, design interaction prototypes for each module, and implement them with consistent placement, wording, and colour so the user can build an accurate mental model across all six modules — demonstrated through **Personal App**, a local-first Flutter application featuring rich-text notes, habit tracking, a calendar with events, a scientific calculator, a life tracker, and customizable settings with onboarding.

2. Objectives

1. Understand the principles of interaction design and functional layout across multi-module applications.
2. Translate user requirements into working design prototypes for each of the six modules (Activity 1).
3. Apply consistency rules — button placement, label wording, colour scheme — across every screen in every module (Activity 2).
4. Evaluate whether the resulting cross-module interface supports an accurate, predictable mental model.

3. Theory

3.1 Interaction Design Across Modules

Interaction design (IxD) focuses on the sequence of screens and states a user moves through to complete a task. In a multi-module app, IxD must ensure that each module has a clear entry point, predictable transitions, and consistent feedback. The Notes module uses a list-to-editor flow, Habits uses a

dashboard-to-detail flow, Calendar uses a grid-to-event flow, Calculator uses a display-to-history flow, and Life Tracker uses an empty-to-dashboard flow.

3.2 Functional Layout

Each module arranges its elements to support its primary task: the Notes FAB creates notes, the Habits FAB adds habits, the Calendar FAB creates events, and the Calculator places digits and operators in a standard grid. The bottom navigation bar or navigation rail persists across all modules with the active tab highlighted.

3.3 Consistency Principles

Four concrete rules are applied across all six modules:

- **Placement consistency** — FABs are always bottom-right. Search bars are top-left. Delete/destructive actions are always accessed via menus or confirmation dialogs.
- **Wording consistency** — “Save”, “Cancel”, “Delete”, “Discard”, “Retry” use identical labels across modules.
- **Colour consistency** — Blue = primary action, red = destructive, amber = warning, green = success. These meanings are preserved in all modules.
- **Icon consistency** — The same icon set and symbol meanings are used throughout (trash = delete, + = create, search = find, etc.).

3.4 Cross-Module Mental Model

When placement, wording, colour, and icons stay consistent across modules, a user who learns the interface in Notes can correctly predict how Habits, Calendar, Calculator, Life Tracker, and Settings will behave. This reduces the learning curve from six independent modules to one transferable pattern.

4. Software / Tools Used

Tool	Purpose
Flutter 3.22+	Cross-platform UI framework (Material 3 design system)
Dart 3.7+	Programming language with sound null safety
Provider	State management (ChangeNotifier-based, 6 providers)
SQLite (sqflite)	Local relational database (notes, habits, events, history, settings)
flutter_quill	Rich text editor for notes (bold, italic, lists, headers, links, images)

Tool	Purpose
flutter_local_notifications + timezone	Push notification scheduling (habits, notes, events)
shared_preferences	Key-value storage for preferences
share_plus / file_picker	Export/import data via JSON files
intl	Date/time formatting across modules
package_info_plus	App version display in Settings

5. Activity 1 — Design and Prototype the Interaction

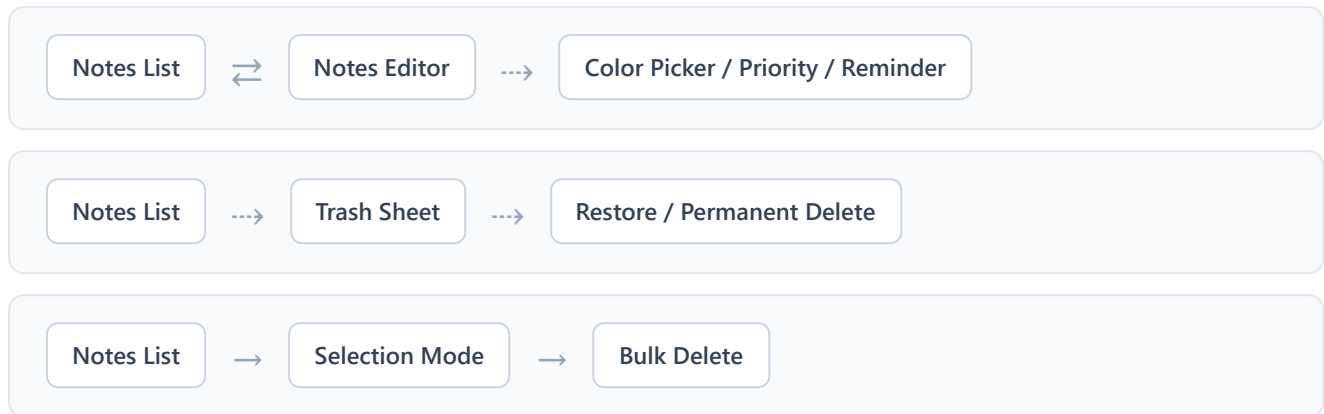
Case study: **Personal App — All Six Modules.**

Module 1 **Notes**

5.1.1 User requirements — Notes

#	User requirement	Screen / interaction
1	Create and organize rich-text notes with formatting.	Notes Editor — Quill rich text
2	Visually distinguish notes by colour.	Color Picker — 14 presets + custom HSV/Hex
3	Set priority levels on notes.	Priority Picker — None/Low/Medium/High
4	Find notes quickly without scrolling.	Search bar — real-time title/content filter
5	Bulk-delete or bulk-move notes to trash.	Selection Mode — long-press, select-all, bulk delete
6	Recover accidentally deleted notes.	Trash bottom sheet — restore or permanent delete
7	Pin or favourite important notes.	Note card — pin icon + favourite star
8	Set reminders on notes.	Editor — reminder time picker + notification
9	See when a note was last updated.	Note card — relative timestamp
10	Confirm before losing unsaved changes.	Discard Changes dialog
11	Feedback when notes cannot be loaded.	Loading spinner + Error state with Retry
12	Clear empty state when no notes exist.	Empty state with illustration + CTA

5.1.2 Screen flow — Notes



5.1.3 Views — Notes

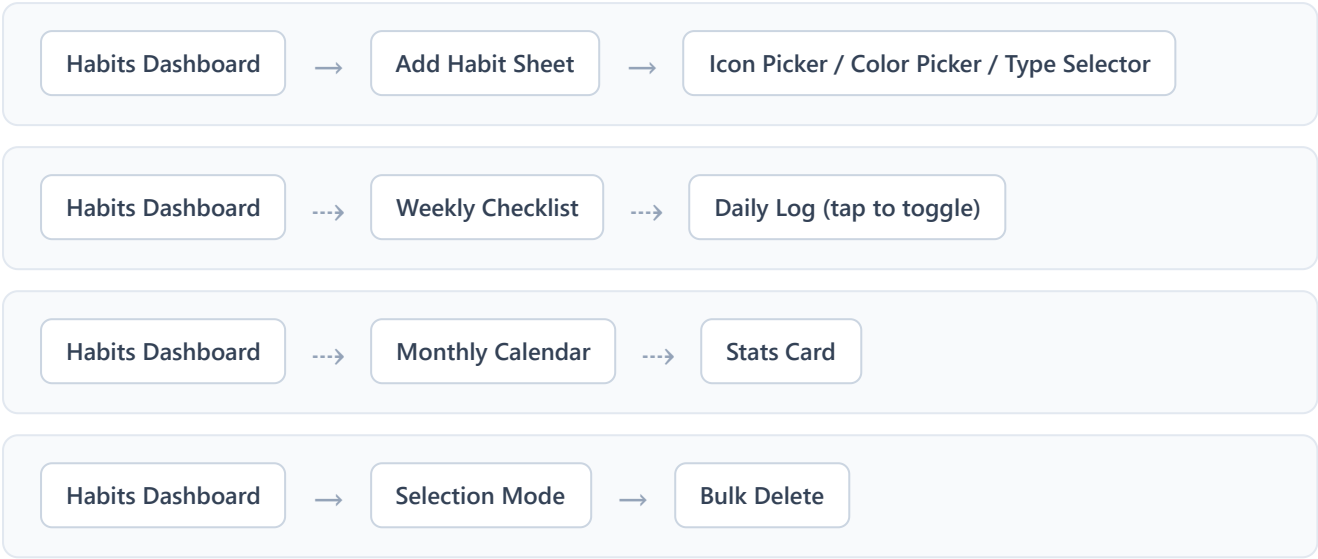
12 views: Loading, Error, Empty, List (data), Search (results), Search (no results), Selection Mode, Editor, Color Picker, Priority Picker, Trash (items), Trash (empty), Discard dialog.

Module 2 Habits

5.2.1 User requirements — Habits

#	User requirement	Screen / interaction
1	Create habits with a name, icon, and colour.	Add Habit bottom sheet — icon picker (55 icons), color picker (12 presets)
2	Choose a tracking type for each habit.	Type selector — Yes/No, Count+Target, Free Count
3	Log daily progress with a single tap.	Weekly checklist — tap to toggle, +/- buttons for count types
4	View weekly progress at a glance.	Weekly checklist row — 7-day horizontal scroll with nav
5	View monthly completion in a calendar grid.	Monthly log calendar — color-coded cells by completion
6	Track habit streaks.	Stats card — current streak + best streak
7	Search habits by name.	Search bar — real-time filtering
8	Reorder habits by dragging.	Horizontal scrollable cards — drag to reorder
9	Set a daily reminder for habits.	Add/Edit habit — time picker + repeating notification
10	Bulk delete habits via multi-select.	Selection Mode — long-press, select-all, delete
11	See today's overall progress.	Today Progress Card — X of Y done with progress bar
12	Get quick suggestions for common habits.	ActionChips — Read, Meditate, Walk, Drink Water

5.2.2 Screen flow — Habits



5.2.3 Views — Habits

13 views: Dashboard, Today Progress Card, Weekly Checklist, Monthly Calendar, Stats Card, Add Habit sheet, Edit Habit sheet, Icon Picker (9 categories), Color Picker, Type Selector, Search (results), Search (no results), Selection Mode, Delete dialog.

Module 3

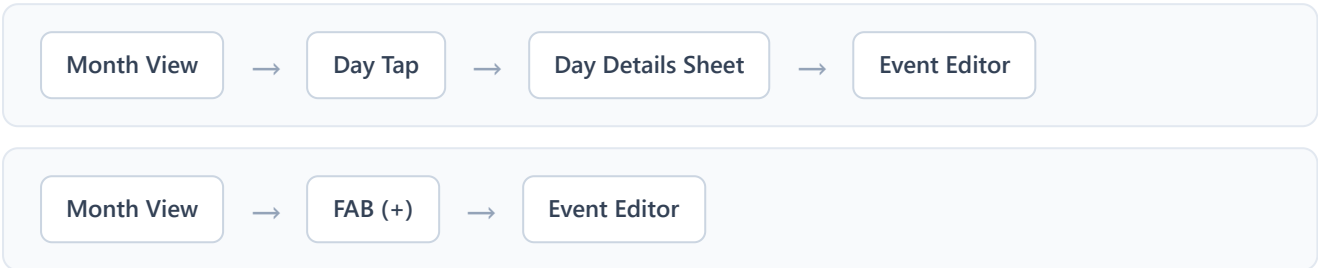
Calendar

5.3.1 User requirements — Calendar

#	User requirement	Screen / interaction
1	View a monthly calendar with navigation.	Main Calendar — swipe/arrows for prev/next month
2	Create events with title, date, time, and category.	Event Editor bottom sheet — form fields + category dropdown
3	See event indicators on calendar days.	Month grid — colored dots on days with events
4	Filter events by category.	Filter chip row — General, Work, Personal, Urgent, All
5	Search events by title or notes.	Search bar — real-time filtering
6	Set event recurrence.	Recurrence dropdown — None/Daily/Weekly/Monthly + end date
7	Receive event notifications.	Auto-scheduled notification at event time
8	Edit or delete events from the day view.	Day details bottom sheet — Edit/Delete buttons
9	Today is always highlighted.	Month grid — circle highlight on current date

#	User requirement	Screen / interaction
10	Confirm before losing unsaved event changes.	Discard Changes dialog

5.3.2 Screen flow — Calendar



5.3.3 Views — Calendar

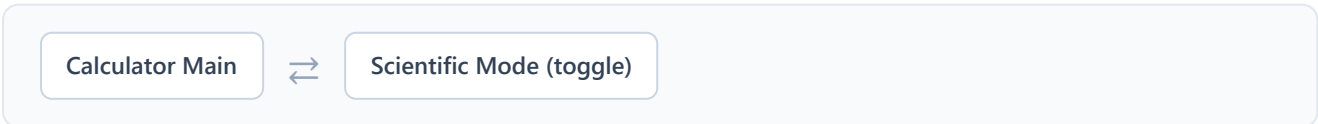
8 views: Month Grid (with events), Month Grid (no events), Day Details sheet (with events), Day Details sheet (empty), Event Editor (create), Event Editor (edit), Search (results), Search (no results), Discard dialog.

Module 4 Calculator

5.4.1 User requirements — Calculator

#	User requirement	Screen / interaction
1	Perform basic arithmetic operations.	Button grid — +, −, ×, ÷
2	Use parentheses for expression grouping.	Button grid — (and) buttons
3	Toggle scientific functions when needed.	SCI toggle — adds sin, cos, tan, log, ln, sqrt, π, e
4	Store and recall values with memory.	Memory row — MC, MR, M+, M- with indicator
5	View and reuse past calculations.	History bottom sheet — last 50 entries, tap to reload
6	Clear or backspace entries.	Clear (C) and backspace buttons
7	Copy result to clipboard.	Optional copy-on-tap setting
8	See clear error messages on invalid input.	Display shows “Error” for invalid expressions
9	Smart number formatting for readability.	Auto-integer, trims trailing zeros, exponential for >15 chars

5.4.2 Screen flow — Calculator



Calculator Main



History Sheet



Tap entry → reload expression

5.4.3 Views — Calculator

5 views: Basic Mode, Scientific Mode, History Sheet (with entries), History Sheet (empty), Error state (display shows “Error”).

Module 5 Life Tracker

5.5.1 User requirements — Life Tracker

#	User requirement	Screen / interaction
1	Set date of birth to begin tracking.	DOB picker dialog — date selector with save
2	Set life expectancy (default 80 years).	Life expectancy dialog — number input 1–120
3	View elapsed time since birth in years/months/days.	Dashboard — elapsed stat block
4	View life progress as a percentage with colour coding.	Progress bar — <50% blue, <80% amber, ≥80% deep orange
5	View expected remaining time.	Dashboard — remaining stat block
6	See real-time metrics (days, weeks, hours, minutes, seconds).	Metric cards — auto-updating every second
7	Reset life tracker data.	Reset confirmation dialog
8	Clear guidance when no DOB is set.	Empty state — illustration + “Set your date of birth” CTA

5.5.2 Screen flow — Life Tracker

Life Tracker (empty)



Set DOB Dialog



Live Dashboard

Live Dashboard



Life Expectancy Dialog



Update / Reset

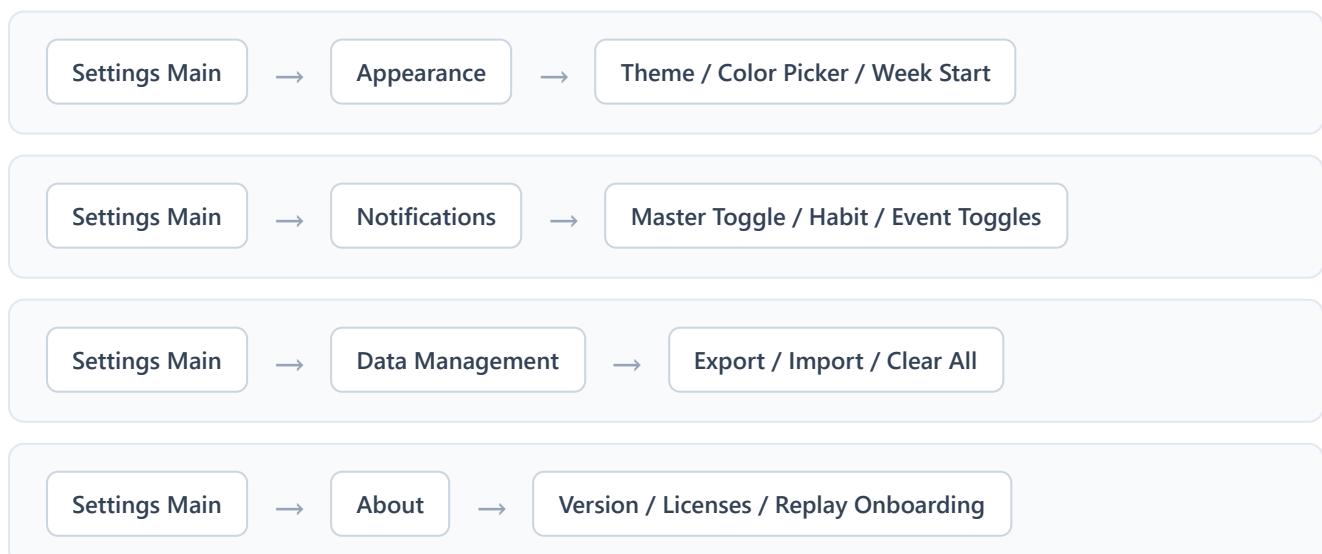
5.5.3 Views — Life Tracker

6 views: Empty State (no DOB), Live Dashboard, DOB Picker dialog, Life Expectancy dialog, Reset confirmation dialog, Error state (DB failure).

5.6.1 User requirements — Settings

#	User requirement	Screen / interaction
1	Switch between Light, Dark, and System theme.	Theme toggle — segmented button, persisted immediately
2	Customize accent colour.	Color picker — 16 presets + custom HSV/Hex with live preview
3	Choose week start day (Monday vs Sunday).	Toggle switch — affects Calendar module
4	Enable or disable all notifications.	Master toggle — requests runtime permission
5	Toggle habit reminders on/off.	Habit reminder toggle
6	Toggle event reminders on/off.	Event reminder toggle
7	Export all data to a JSON file.	Export button — generates JSON, opens system share sheet
8	Import data from a JSON backup.	Import button — file picker, validates + DB transaction
9	Clear all app data with confirmation.	Clear button — two-step confirmation, must type “DELETE”
10	View app version and open source licenses.	About section — version, GitHub link, licenses page
11	Replay the onboarding tutorial.	Onboarding replay link — resets SharedPreferences flag
12	Configure calculator preferences.	Calculator section — scientific mode toggle, copy-on-tap toggle, clear history

5.6.2 Screen flow — Settings



5.6.3 Views — Settings

12 views: Main Settings (scrollable), Appearance section, Color Picker (16 presets), Custom Color (HSV/Hex dialog), Notifications section, Data Management section, Import (file picker), Import (loading), Import (success/error), Clear All (two-step confirmation), About section, Licenses page.

Supporting

Onboarding

5.7.1 User requirements — Onboarding

#	User requirement	Screen / interaction
1	See a visual introduction to all app features.	5-page PageView — Notes, Habits, Calendar, Calculator, Life Tracker
2	Skip onboarding at any time.	Skip button — visible on pages 1–4
3	Navigate between pages with dot indicators.	Dot indicator row + Next button
4	Only see onboarding once.	SharedPreferences flag — checked in main.dart

5.7.2 Screen flow — Onboarding



5.7.3 Views — Onboarding

5 views: 5 onboarding pages + Skip/Get Started buttons.

6. Activity 2 — Apply and Verify Consistency

Consistency rules were applied once at the app level (via Material 3 theme, shared widgets, and a common design system) and reused across all six modules rather than implemented independently per module.

6.1 Consistency rules applied

Button Placement

FAB is always bottom-right across Notes, Habits, and Calendar. Delete actions are always accessed via confirmation dialogs. Save is always top-right. Cancel is always left, Confirm is always right in dialogs. Consistent across all 6 modules.

Label Wording

'Save', 'Cancel', 'Delete', 'Discard', 'Retry', 'Apply' never change across modules. Confirmation dialogs echo the verb: 'Delete habit?', 'Clear all data?'. Consistent error messages use 'Something went wrong' with Retry.

Colour Scheme

 **Blue** — primary action, FAB, active tab, save
 **Red** — destructive (delete, discard, high priority)
 **Green** — success, completion, low priority
 **Amber** — warning, medium priority, caution
No colour is reused for a different meaning across modules.

Icon & Navigation Consistency

Search is always the leftmost icon in app bars. The create FAB is always bottom-right. The bottom NavigationBar (or NavigationRail on wide screens) uses an IndexedStack to preserve state. Icons use the same symbol set throughout (trash, +, back, search, settings gear).

6.2 How this builds a cross-module mental model

Because the same rules are reused across all six modules, the user needs to learn the interaction pattern only once. After using Notes, the user already knows:

- The blue FAB creates new content (note, habit, event).
- A red button means deletion (always confirmed via dialog).
- Long-press opens selection mode for bulk actions.
- Search always works the same way — real-time filtering at the top.

- Theme and colour choices in Settings instantly apply to all modules.
- The back arrow or swipe gesture always navigates to the previous screen.

Cross-Module Consistency Matrix

Rule	Notes	Habits	Calendar	Calculator	Life Tracker	Settings
FAB bottom-right	✓	✓	✓	N/A	N/A	N/A
Search top-left	✓	✓	✓	N/A	N/A	N/A
Blue = primary action	✓	✓	✓	✓	✓	✓
Red = destructive	✓	✓	✓	N/A	✓	✓
Confirm before delete	✓	✓	✓	N/A	✓	✓
Same wording for actions	✓	✓	✓	✓	✓	✓
Error/Loading/Empty states	✓	✓	✓	✓	✓	✓
Shared theme applies	✓	✓	✓	✓	✓	✓

7. Output / Result

The completed Personal App prototype implements all six modules within a single Flutter application sharing a common Material 3 design system. The app opens with a 5-page onboarding flow (first launch only), then presents six tabs via IndexedStack: Notes (rich text with colour/priority/trash), Habits (three tracking types with streaks), Calendar (month grid with events and recurrence), Calculator (basic + scientific with memory), Life Tracker (real-time dashboard), and Settings (appearance, notifications, data management).

The design system is configured once via Material 3’s theme system with a user-selectable color scheme seed, and every screen inherits the same placement, wording, and colour rules automatically.

7.1 Self-evaluation checklist

✓ **Is the primary action in the same position across all modules?**

Yes — FAB bottom-right in Notes, Habits, Calendar; Calculator uses display area consistently

✓ **Does the same action always use the same word?**

Yes — Save / Cancel / Delete / Discard / Retry / Apply identical across all 6 modules

✓ **Is each colour assigned exactly one meaning?**

Yes — Blue = action, Red = destructive, Green = success, Amber = warning — consistent in all modules

✓ **Can a user predict an unseen module from a seen one?**

Yes — Habits reuses FAB + search + selection patterns from Notes; Settings reuses dialog patterns

✓ **Does every loading/error state provide feedback and recovery?**

Yes — Loading spinner + Error state with Retry button in Notes, Habits, Calendar, Settings import

✓ **Is every destructive action protected by a confirmation dialog?**

Yes — Delete, Discard, Clear All Data, Reset Life Tracker all require explicit confirmation

✓ **Are both empty and populated states handled everywhere?**

Yes — Notes, Habits, Calendar, Life Tracker, Calculator history all have empty states with CTAs

✓ **Does the navigation persist across all modules?**

Yes — IndexedStack preserves all 6 tabs; NavigationRail on wide screens, NavigationBar on narrow

Annexure A — Selected UI/UX Ideas Integrated

Seven ideas from the reference design-thinking framework were integrated across the Personal App modules:

1. Progressive Onboarding with Skip

Module: Onboarding | 5-page PageView introducing all features. Skip button visible on pages 1–4. Only shows on first launch via SharedPreferences flag. The Notes page introduces rich text editing before the user even opens the app.

2. Colour + Priority Visual Encoding

Module: Notes + Habits | Both modules use colour to convey meaning. Notes uses 14 colour presets + custom for visual distinction and 4 priority levels. Habits uses 12 colour presets per habit. Both use a shared custom HSV/Hex picker component.

3. Search with Real-Time Feedback

Modules: Notes, Habits, Calendar | All three modules implement the same search pattern: debounced text input at the top of the screen, real-time filtering, and both hit and no-result states with clear messaging.

4. Trash with Restore Safety Net

Module: Notes | Soft delete model with a dedicated Trash bottom sheet. Users restore individual items or permanently delete. Clear timestamps show deletion date. "Empty Trash" bulk action available.

5. Streak Tracking for Habit Motivation

Module: Habits | Current streak and best streak calculated from consecutive daily completions. Displayed in the Stats Card alongside the monthly completion percentage and progress bar. Drives daily engagement through the "don't break the chain" effect.

6. Live Real-Time Counter

Module: Life Tracker | Uses a Stream.periodic timer to update elapsed time every second. All metrics (years, months, days, hours, minutes, seconds) update in real time on the dashboard, creating a compelling "clock" effect that encourages user return.

7. Live Theme Preview with Custom Color Picker

Module: Settings | The color seed picker shows a live SV gradient area with a hue slider and Hex input. The selected colour applies instantly to the Material 3 theme via colorSchemeSeed. Users see their changes in real time before committing.

Annexure B — Screen Wireframes (Complete Application)

Eleven representative screens from all six modules plus onboarding, showing the full application user journey.

